

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Technova Official Website

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Prepared By:

Club Mentor: Anmol Mahanty

Development Team:

- **Anmol Mahanty (project lead)**
- **Harshali Wani**

Design Support: Pranay Shah

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements of the TechNova Official Website. The platform shall serve as the official digital system for managing events, announcements, resources, attendance, feedback, and student engagement activities for the TechNova club.

1.2 Scope

The TechNova Website shall provide:

- A public-facing club website
- A student portal for event participation
- An admin dashboard for operational control

The system shall support event discovery, registration, attendance tracking, feedback collection, announcements, and long-term event showcasing.

2. Overall Description

2.1 Product Perspective

The TechNova Website is a web-based application consisting of:

- A public landing interface
- Authenticated student features
- Role-restricted admin operations

The system is designed to be scalable, secure, and cost-effective using cloud services.

2.2 Product Functions

- Club landing pages (Home, About, Team, Contact)
- Event listing and management
- Student event registration
- Announcements with optional email notifications
- Event highlights gallery
- Event resource distribution
- QR-based attendance management
- Feedback collection and analytics

2.3 User Classes and Characteristics

- **Public Users:** Can view public content without authentication.
- **Student Users:** Must authenticate to register for events, access resources, mark attendance, and submit feedback.
- **Admin Users:** Privileged users assigned manually who manage all operational activities.

2.4 Operating Environment

- Web application compatible with modern browsers
- Optimized for desktop and mobile devices

2.5 Constraints

- Must operate within limited budget constraints
- Must use secure authentication and authorization
- Must follow phased deployment

2.6 Assumptions & Dependencies

- Firebase and Vercel services remain available
- Admin credentials are managed internally by TechNova

3. System Architecture Overview

3.1 Technology Stack

Component	Technology	Purpose
Frontend	React (Vite)	UI rendering
Backend	Node.js + Express (Vercel Functions)	APIs
Authentication	Firebase Auth	Role-based login
Database	Firebase Firestore	Events, sessions, registrations, attendance, feedback
Storage	Cloudinary	Media & resources
Email	Resend	Notifications
Background Jobs	Firebase Cloud Functions	Cleanup & scheduled tasks

4. Functional Requirements

This section provides a high-level summary. Detailed behavior is defined in Sections 4.1–4.8.

FR-1 Landing Website

The system shall display a responsive landing website with navigation to Home, About, Events, Team, and Contact pages.

FR-2 Authentication & Authorization

The system shall support student authentication via email/password and Google Sign-In. The system shall restrict admin access using role-based authorization.

FR-3 Events Module

The system shall display events categorized as Upcoming, Ongoing, or Past based on admin-defined status.

The system shall support events with one or more sessions.

FR-4 Announcements

The system shall allow admins to publish announcements and optionally send email notifications.

FR-5 Event Highlights

The system shall allow admins to manage event highlight media visible to all users.

FR-6 Resources

The system shall restrict event resources to registered students only.

FR-7 Attendance

The system shall generate QR codes per student per session and store session-wise attendance.

FR-8 Feedback

The system shall allow feedback submission only after attendance completion for all sessions.

4.1 Authentication & Authorization Flow

4.1.1 User Roles

The system shall support the following user roles:

- **Student:** Default role assigned to all newly registered users.
- **Admin:** Privileged role assigned manually to selected users by TechNova authorities.

Role information shall be stored in the database and enforced across the system through role-based access control.

4.1.2 Student Registration Flow

The system shall allow students to register using either of the following methods:

A. Email and Password Registration

The registration form shall collect the following mandatory details:

- First Name
- Last Name
- Phone Number
- Email Address
- Password
- Confirm Password

The system shall validate that:

- The email address is unique.
- The password and confirm password fields match.
- All mandatory fields are provided.

Upon successful registration:

- A student account shall be created using Firebase Authentication.
- The user shall be assigned the default role of **Student**.
- The user profile data shall be stored in the database.
- The user shall be redirected to the Student Dashboard.

B. Google Sign-In Registration

The system shall support student registration using Google Sign-In via Firebase Authentication.

Upon successful Google authentication:

- A student account shall be created automatically.
- The user shall be assigned the default role of **Student**.
- Available profile information shall be stored in the database.
- The user shall be redirected to the Student Dashboard.

4.1.3 Student Login Flow

The system shall allow students to log in using either:

- Email and Password, or
- Google Sign-In via Firebase Authentication.

Upon successful login:

- The system shall verify the user's role.
- Students shall be redirected to the Student Dashboard.

If login credentials are invalid:

- The system shall display the error message "**Invalid credentials**".
- No sensitive information shall be revealed regarding the cause of the failure.

4.1.4 Admin Login Flow

The system shall not provide a separate admin registration or login interface.

Admin authentication shall follow these rules:

- Admin users shall log in using the same login interface as students.
- Admin privileges shall be granted manually by assigning the role **Admin** to predefined email addresses.
- Role verification shall occur immediately after successful authentication.

Upon successful login:

- Users with the **Admin** role shall be redirected to the Admin Dashboard.
- Users without the **Admin** role attempting to access admin routes shall be redirected to the Home page.

4.1.5 Authorization & Route Protection

The system shall enforce strict role-based authorization:

- Public routes shall be accessible without authentication.
- Student routes shall require authentication.
- Admin routes shall be accessible only to users with the Admin role.

Unauthorized access attempts to admin routes shall result in automatic redirection to the Home page without exposing administrative content.

Authorization rules shall be enforced at both:

- User interface level, and
- Backend/database access level.

4.2 Events Management & Visibility Flow

4.2.1 Event Creation & Management (Admin Perspective)

The system shall provide an Admin-only Event Management module.

Admins shall create events with:

- Name
- Description
- Event duration (for display only)
- Registration deadline
- Status
- Capacity
- One primary media asset

Event duration shall not govern attendance logic.

The system shall allow admins to:

- Save events as **Draft** before publishing.
- Publish events directly without saving as draft.
- Edit any event details at any time.

- Unpublish events when required.

If an admin attempts to unpublish or modify a published event:

- The system shall display a warning confirmation popup before applying changes.

4.2.2 Event Status, Capacity & Registration Control

Event status shall be **manually controlled by admins only**.

The system shall support the following rules:

- Events marked as **Upcoming** may have registrations enabled or disabled.
- Event registration capacity shall be configurable by admin.
- If the registration capacity is reached, further registrations shall not be allowed.
- Events marked as **Past** shall not allow registrations.
- Registration availability shall be manually controlled by admins.

If registration is closed:

- The system shall display the message
“**Registration Closed. Contact Admin.**”
- Admins shall be able to configure a contact phone number to be displayed with the message.

4.2.3 Event Media & Highlights

Each event shall support:

- **Only one primary media asset** (image or video) for event detail showcase.

After an event is completed:

- Admins may upload multiple images as **Event Highlights**.
- Event highlights shall be visible to all users (registered and unregistered).

Event highlights are considered permanent branding content and are not subject to automatic data cleanup policies.

4.2.4 Event Resources Management

Admins shall be able to add event-specific resources including:

- PDF files
- PPT files
- External links

Resources shall be associated with a specific event and managed only by admins.

4.2.5 Event Visibility — Unregistered Users

Users who are not logged in shall:

- Be able to view event details for both upcoming and past events.
- Not be allowed to register for any event.

The system shall prompt unregistered users to log in or register when attempting to register for an event.

4.2.6 Event Visibility — Registered Students

Authenticated student users shall:

- Be able to view event details for both upcoming and past events.
- Be able to register for upcoming events when registration is active and capacity is available.
- Be able to view and download event resources only if registered for that event.

Access to event resources shall be strictly restricted to students registered for the corresponding event.

4.2.7 Access Enforcement

Event visibility and actions shall be enforced through:

- UI-level access control
- Backend and database-level authorization rules

No event registration or resource access shall be permitted without proper authentication and authorization.

4.2.8 Event Deletion Policy

The system shall allow admins to permanently delete events.

The following rules shall apply:

- Only admins shall have permission to delete events.
- Event deletion shall be irreversible.
- Before deleting an event, the system shall display a confirmation popup warning the admin about permanent data removal.
- Upon deletion, all associated event data including registrations, resources, attendance records, feedback, and highlights shall also be removed from the system.
- This deletion policy applies only when an Admin manually deletes an event. Automatic data retention and cleanup rules defined in Section 4.5.10 apply only to completed events that are not manually deleted

The delete option shall be available alongside publish and unpublish actions in the admin event management interface.

Manual event deletion bypasses the 30-day retention policy and results in immediate permanent removal of all associated data.

4.2.9 Event Sessions Management

The system shall support events consisting of one or more sessions conducted across one or more days.

Each session shall include:

- Session Name
- Session Date
- Session Start Time
- Session End Time

Admins shall be able to add, modify, or delete sessions before or during an event. Sessions added during an ongoing event shall immediately affect attendance, feedback eligibility, statistics, and exports.

4.3 Event Registration Flow

4.3.1 Registration Eligibility

The system shall allow event registration only under the following conditions:

- The user must be authenticated as a Student.
- The event must be marked as **Upcoming**.
- Event registration must be enabled by the Admin.
- The current date and time must be before the event's registration last date and time.
- Event registration capacity must not be exceeded.

If any of the above conditions are not met, event registration shall not be permitted.

4.3.2 Registration Form & Additional Questions

During event registration, the system shall require students to answer additional questions configured by the Admin for each event.

Examples of such questions may include (but are not limited to):

- Academic Year
- Branch / Department
- College Name

The set of questions shall be configurable by the Admin at the time of event creation or modification.

Student responses shall be stored along with the event registration data.

4.3.3 Registration Submission & Success Handling

Upon successful submission of the registration form:

- The system shall immediately register the student for the event.
- An **instant success message** shall be displayed on the same event page.
- The **Register** button shall be removed.
- A **Resources** button shall be displayed in place of the Register button.

No intermediate confirmation page or modal shall be shown.

4.3.4 Post-Registration Event Page Behavior

The system shall dynamically update the event page based on the student's registration status:

- If the student **is registered** for the event:
 - The Register button shall not be visible.
 - A **Resources** button shall be displayed.
 - Clicking the Resources button shall redirect the user to the event's resource page.
- If the student **is not registered** for the event:
 - The Register button shall be visible (subject to registration availability).

This behavior shall apply consistently across upcoming and past events.

4.3.5 Duplicate Registration Prevention

The system shall prevent duplicate registrations for the same event.

If a registered student revisits the event page:

- The system shall not allow re-registration.
- The event page shall display the **Resources** button instead of the Register button.

No duplicate registration records shall be created.

4.3.6 Email Confirmation

After successful event registration:

- The system shall send a **basic confirmation email** to the student.
- The email subject and content shall be configurable by the Admin during event creation or modification.

The email shall serve as confirmation of successful registration.

4.3.7 Registration Cancellation Policy

The system shall not provide any option for students to cancel or unregister from an event.

Only Admins shall have the authority to remove or manage registrations if required.

4.3.8 Automatic Registration Closure

The system shall automatically close event registrations after the configured **registration last date and time**, regardless of manual admin action.

Once registration is closed:

- No further registrations shall be permitted.
- The system shall display the message:
 “Registration Closed. Contact Admin”, along with the admin-configured contact number.

4.4 Event Resources Access Flow

4.4.1 Resource Ownership & Management

The system shall support **event-wise resource management**.

The following rules shall apply:

- Only Admins shall be permitted to upload, modify, or delete event resources.
- Resources shall be associated with a specific event.
- Supported resource types shall include:
 - PDF files
 - PPT files
 - External links

4.4.2 Resource Enablement & Availability

Event resources shall not be visible by default.

The system shall enforce the following behavior:

- Event resources shall become accessible to students **only after the Admin manually enables and adds resources** for the event.
 - Registration alone shall not automatically unlock access to resources.
 - Admins shall have full control over when resources are made available.
-

4.4.3 Resource Visibility & Access Control

Resource access shall be governed by strict authorization rules:

- Only students who are **registered for the corresponding event** shall be allowed to access event resources.
 - Unregistered users and unregistered logged-in students shall not be able to view or access event resources.
 - Registered students shall retain access to event resources even after the event is marked as **Past**.
-

4.4.4 Resource Viewing & Downloading

Registered students shall be allowed to:

- View resources directly on the platform where applicable.
- Download resources freely without restrictions.

No download limitations shall be imposed once resource access is enabled by the Admin.

4.4.5 Resource Updates

If Admins upload new resources or modify existing resources at any time:

- All registered students shall automatically be able to view the newly added or updated resources.
 - No additional action from the student shall be required.
-

4.4.6 Resource Deletion Policy

Admins shall be able to delete event resources.

The following rules shall apply:

- Resource deletion shall be **permanent and immediate**.
 - Deleted resources shall not be recoverable.
 - Once deleted, the resource shall no longer be visible to any user.
-

4.4.7 Resource Display Order

Event resources shall be displayed to students in the **order in which they were uploaded** by the Admin.

No manual reordering or grouping shall be required.

4.5 Attendance Management (QR-Based) Flow

4.5.1 Attendance Eligibility

The system shall allow attendance marking only for registered students and only once per session.

4.5.2 QR Code Generation

The system shall generate a unique QR code per student per session containing:

- Event ID
 - Session ID
 - Student ID
-

4.5.3 Attendance Activation Window

- Attendance window is now **per session**
 - For each session:
 - Opens **1 hour before session start**
 - Closes **1 hour after session start**
 - Admin can manually open/close attendance for any session
-

4.5.4 Attendance Scanning Process (Admin-Controlled)

Attendance scanning shall be performed by **Admins only**.

The process shall be as follows:

1. The student presents their attendance QR code.
2. The Admin scans the student's QR code.
3. The system validates:
 - Student registration status
 - Event ID match
 - Whether attendance has already been marked

Upon successful scan:

- Attendance shall be marked for the student.
- The Admin shall receive a confirmation popup displaying the student's name and check-in status.

4.5.5 Manual Attendance Override

To handle QR failures or technical issues:

- Admins shall be able to **manually mark attendance** for a registered student.
- Manual attendance marking shall follow the same validation rules as QR-based attendance.
- Manual overrides shall be logged as admin-initiated actions.

4.5.6 Duplicate Attendance Prevention

The system shall prevent duplicate attendance entries.

If an attempt is made to mark attendance for a student who has already been marked present:

- The system shall block the action.
- The Admin shall be notified that attendance has already been recorded.

4.5.7 Post-Attendance Student Experience

After successful attendance marking:

- The attendance QR code button shall be removed or disabled.
- A **success message** shall be displayed.
- The **Feedback** button shall be unlocked and displayed in place of the attendance button.

- Feedback unlocks only after the student has successfully marked attendance for all sessions of the event, as defined in Section 4.6.

Students shall not be able to access feedback without successful attendance marking.

4.5.8 Attendance Data Access

Attendance records shall:

- Attendance records shall be stored **session-wise**, linked to events and student users.
- Be accessible **only to Admins** for viewing and analysis.
- Not be editable by students.

4.5.9 Statistics & Export

Admins shall view:

- Event-level statistics
- Session-wise attendance breakdown

Exports shall include session-wise attendance data suitable for certificates and reporting.

4.5.10 Data Retention Policy

Operational event data shall be retained for 30 days after event completion and then permanently deleted.

Event highlights and metadata shall not be deleted automatically.

4.5.9 Event Attendance & Registration Statistics (Admin View)

4.5.9.1 Event-Wise Statistics Overview

For each event, the system shall provide Admins with an event-wise statistics dashboard.

The dashboard shall display the following metrics:

- Total number of **registered students**
- Total number of **checked-in (attended) students**
- Number of registered students who **did not check in**
- Attendance completion status per event

For the purpose of this document, ‘checked-in’, ‘attended’, and ‘completed attendance’ refer to students who have successfully marked attendance for all sessions of an event.”

These statistics shall update in real time as registrations and attendance records change.

4.5.9.2 Registered Students List

Admins shall be able to view a complete list of all students registered for a specific event.

The registered students list shall include:

- Student Full Name
- Email Address
- Phone Number
- Responses to admin-defined registration questions (e.g., year, branch, college)
- Registration timestamp

This list shall be displayed separately from attendance data.

4.5.9.3 Checked-In Students List

Admins shall be able to view a separate list of students who have successfully marked attendance **for all sessions of the event**.

The checked-in students list shall include:

- Student Full Name
- Email Address
- Attendance status (Checked-in)
- Attendance timestamp
- Attendance method (QR-based or Manual override)

4.5.9.4 Data Separation & Comparison

The system shall clearly distinguish between:

- Registered students
- Checked-in (attended) students

Admins shall be able to:

- Identify students who registered but did not attend
- Compare registration and attendance counts easily

4.5.9.5 Data Export & Download

Admins shall be able to download event-wise data for future use.

The system shall support downloading:

- Registered students list
- Checked-in students list

The exported data shall be suitable for documentation, certificate generation, and reporting purposes.

The export functionality shall preserve:

- Student details
- Event details
- Registration and attendance timestamps
- Admin-defined registration responses

EXPORT STRUCTURE EXTENDED

Downloaded data now includes:

- Session Name
- Session Date
- Session-wise attendance status per student

This explicitly supports:

- Certificate generation
 - Documentation
 - Audit reports
-

4.5.9.6 Access Control

- Only Admins shall be permitted to view or download event statistics and student lists.
- Students shall not have access to any registration or attendance statistics.

SESSION-WISE STATISTICS ADDED

Admins can now see:

- **Event-level stats**
 - Total registrations
 - Total students who completed all sessions
- **Session-level stats**
 - Session name & date
 - Attendance count per session
 - Who attended which session

4.5.10 Event Data Retention & Automatic Cleanup Policy

After an event is completed, the system shall support controlled data retention and automatic cleanup of operational data.

The following rules shall apply:

- Admins shall be allowed to **download all event-related operational data** after the event ends.
- Downloadable data shall include:
 - Event details
 - Event registration records
 - Attendance records
 - Feedback entries
 - Registration form responses

Downloaded data shall be suitable for:

- Documentation
- Certificate generation
- Reporting purposes

After the event date:

- The system shall retain all event-related operational data for **30 days**.
- Upon completion of 30 days from the event date, the system shall **automatically delete selected event-specific operational data**.

The following data shall be **permanently deleted** after 30 days:

- Event registration records
- Attendance records
- Feedback entries
- Event resources (PDFs, PPTs, links)

The following data shall **NOT** be deleted automatically:

- Event highlights (images and media)
- Event metadata (basic event information)
- Student user accounts
- Admin user accounts
- Global system configuration data

Event highlights shall remain available indefinitely until:

- An Admin manually deletes them, or
- An Admin modifies or replaces them.

Automatic deletion shall be irreversible and shall not require manual admin intervention.

CLEANUP SCOPE CLARIFIED & REFINED

Data that WILL be auto-deleted after 30 days:

- Event registration records
- **Session-wise attendance records**
- Feedback entries
- Event resources (PDF, PPT, links)

Data that WILL NOT be deleted:

- Event highlights (images/media)
- Event metadata
- User accounts (students/admins)

This clarification was **added after correction**.

Dynamic Session Management Rule

Admins can:

- Add sessions
- Add new days
- Do this **even while the event is ongoing**

System must:

- Immediately enforce attendance
- Immediately update feedback conditions
- Immediately reflect in stats & exports

This rule affects:

- Events module
- Attendance module
- Feedback module
- Admin dashboard
- Export logic

★ 4.6 Feedback Management Flow

4.6.1 Feedback Eligibility & Unlocking

The system shall allow feedback submission only under the following conditions:

- The student must be **registered** for the event.
- The student must have **successfully marked attendance** for the event.

Feedback shall become available immediately **after the student has successfully marked attendance for all sessions of the event**

Students who have not marked attendance shall not be able to access or submit feedback.

4.6.2 Feedback Structure

The feedback form shall collect the following inputs:

- **Event Rating** (numerical rating, e.g., 1–5)

- **Speaker Rating** (numerical rating, e.g., 1–5)
- **Comments** (free-text feedback)

The feedback structure shall be **event-wise** and configurable by the Admin.

4.6.3 Feedback Submission Rules

The system shall enforce the following rules:

- Each student shall be allowed to submit **only one feedback entry per event**.
- Once submitted, feedback shall not be duplicated.
- Students shall be allowed to **edit their submitted feedback** until feedback is disabled by the Admin for that event.

If admin adds a new session during an ongoing event:

- Students must attend the **new session as well**
- Otherwise feedback remains locked

When feedback is disabled by the Admin:

- Students shall no longer be able to submit or edit feedback.
 - Previously submitted feedback shall remain stored and visible to Admins.
-

4.6.4 Feedback Edit Control

Admins shall have the ability to:

- Enable or disable feedback for each event.
- Control the time window during which students are allowed to edit their feedback.

Students shall not be able to edit feedback once the Admin disables feedback for the event.

4.6.5 Feedback Visibility & Identity

Feedback shall be:

- **Non-anonymous**
- Visible to Admins along with the **student's identity**

Admins shall be able to view:

- Student name
- Event rating
- Speaker rating
- Comments
- Feedback submission and update timestamps

Feedback shall not be visible to other students.

4.6.6 Admin Feedback View & Analytics

Admins shall have access to both:

- **Raw feedback data**, and
- **Aggregated feedback analytics**

The Admin interface shall provide:

- Average event rating
- Average speaker rating
- Distribution of ratings
- Access to individual feedback entries with student details

These insights shall be available on an event-wise basis.

4.6.7 Data Integrity & Access Control

The system shall ensure:

- Feedback data can be created and modified only by the student who submitted it (within the allowed edit window).
- Feedback data can be viewed only by Admins.
- Feedback data cannot be deleted or altered by students once feedback is disabled.

4.7 Announcements & Email Notification Flow

4.7.1 Announcement Types

The system shall support the following types of announcements:

- **General Announcements:** Applicable to the entire TechNova community.

- **Event-Specific Announcements:** Applicable to a particular event.

Each announcement shall be created and managed by Admins only.

4.7.2 Announcement Creation & Recipient Selection

While creating an announcement, Admins shall be able to:

- Define whether the announcement is **general** or **event-specific**.
- Select the **recipient group** explicitly.

Recipient selection shall be controlled by the Admin and may include:

- All registered users, or
- Users registered for a specific event, or
- Any specific subset supported by the system.

The system shall not automatically determine recipients without Admin selection.

4.7.3 Website Visibility of Announcements

Announcements shall be visible on a **dedicated Announcements page**.

The following visibility rules shall apply:

- Only **registered and logged-in users** shall be able to view announcements.
- Announcements shall not be visible to unregistered or public users.
- No announcements shall be displayed on the Home page.

4.7.4 Email Notification Control

Email notifications for announcements shall be **explicitly controlled by Admins**.

The system shall enforce the following rules:

- Admins must manually enable a **“Send Email”** option while publishing an announcement.
- Email notifications shall not be sent by default.
- Admins may choose to disable email notifications for any announcement.

Each announcement shall be evaluated independently for email sending.

4.7.5 Email Content Format

Announcement emails shall be sent in **plain text format only**.

The email content shall:

- Be defined by the Admin during announcement creation.
- Contain no rich HTML formatting or embedded media.
- Be concise and readable across all email clients.

4.7.6 Announcement Lifecycle

Announcements shall support the following lifecycle states:

- **Published**
- **Unpublished**

The system shall not support draft or archive states for announcements.

Admins shall be able to:

- Publish announcements
- Unpublish announcements at any time

4.7.7 Editing Published Announcements

Admins shall be allowed to edit announcement content after publishing.

The following rules shall apply:

- Editing an announcement shall update its content on the website.
- Editing an announcement shall **not automatically resend email notifications**.
- Email notifications shall be sent only at the time of initial publishing, if enabled.

4.7.8 Access Control & Security

The system shall ensure that:

- Only Admins can create, edit, publish, or unpublish announcements.

- Only registered users can view announcements on the website.
 - Announcement emails are sent only to Admin-selected recipients.
-

4.8.1 Overall Design Philosophy

The TechNova website shall follow a **modern, minimal design approach with subtle galaxy-inspired accents**.

The design shall prioritize:

- Clarity
- Readability
- Performance
- Consistency across pages

The galaxy theme shall enhance branding without overwhelming usability.

4.8.2 Galaxy Theme Usage

Galaxy-inspired visuals shall be applied selectively.

The following rules shall apply:

- Galaxy accents may be used on:
 - Public pages
 - Event pages
- Galaxy visuals shall **not** be used heavily on dashboards.
- Admin interfaces shall avoid galaxy visuals entirely.

Galaxy elements may include:

- Dark space-inspired backgrounds
- Subtle gradients
- Minimal star or nebula accents

Heavy background animations or particle effects shall not be used.

4.8.3 Animation & Motion Policy

The system shall support **micro-animations only**.

Allowed animations include:

- Button hover effects
- Page transitions
- Subtle loading indicators

The system shall not use:

- Background motion animations
- Parallax effects
- Performance-heavy visual effects

All animations shall be lightweight and non-distracting.

4.8.4 Public vs Student UI Behavior

Student-facing pages shall be visually **cleaner and simpler** than public-facing pages.

The following distinctions shall apply:

- Public pages may emphasize branding and visual storytelling.
- Student pages shall prioritize task completion (registration, resources, attendance, feedback).
- Visual effects on student pages shall be minimal.

Navigation and layout consistency shall be maintained across both views.

4.8.5 Admin Interface Design

The Admin dashboard shall follow a **neutral, utility-focused theme**.

The Admin UI shall:

- Avoid galaxy visuals
- Prioritize data density and clarity
- Use clean layouts, tables, and cards

A neutral dark or light theme may be used for Admin pages.

4.8.6 Mobile-First Design Priority

The system shall follow a **mobile-first design approach**.

The UI shall be optimized primarily for:

- Mobile devices (phones)
- Tablets

Desktop layouts shall scale gracefully from the mobile-first base.

All interactive elements shall be designed to be touch-friendly.

4.8.7 Error States & Empty States

Error states and empty states shall be **branded but minimal**.

The system shall:

- Clearly communicate the issue or empty condition
- Avoid playful illustrations or mascots
- Maintain a professional and clean appearance

Messages shall be concise and actionable where applicable.

4.8.8 Accessibility Standards

The system shall adhere to baseline accessibility standards.

The UI shall enforce:

- High contrast between text and background
- Large and accessible tap targets
- Keyboard navigation support

Accessibility shall be considered a core requirement, not an optional enhancement.

5. Non-Functional Requirements

5.1 Performance

- Website pages should load within 2–3 seconds on average campus network.
- API responses should be under 2 seconds for standard operations.

5.2 Security

- Admin pages must be protected via authentication and role-based authorization.
- Firestore security rules must prevent unauthorized access to sensitive collections.
- All communications must use HTTPS.

5.3 Reliability

- The system shall maintain high availability (target 99% uptime) during event periods.

5.4 Scalability

- The system should support at least 2000 student users without performance degradation.

5.5 Usability

- UI should follow modern minimal design.
- Navigation must be intuitive with clear call-to-actions.

6. Budget & Grant Usage Plan (₹5,000)

6.1 Cost Overview

The website architecture is designed to operate largely on free-tier services including Vercel hosting, Firebase authentication and databases, Cloudinary storage, and Resend email service. The primary fixed cost is the domain purchase.

6.2 Grant Usage Plan

A grant of ₹5,000 has been applied for from the college to support website deployment. The grant will primarily be used for the domain purchase. Remaining funds will be treated as contingency and may be utilized only if required for operational expenses such as payment gateway charges, additional email limits, or other service overages.

Item	Purpose	Planned Usage
Domain purchase (3 years)	Official website identity	Primary
Service overages / email limit increase	Contingency if usage exceeds free tier	Contingency
Other required expenses	Any future unforeseen operational needs	Contingency

7. Acceptance Criteria

- Website is deployed live with stable performance.
- Students can register/login and register for events successfully.
- Admins can login and create/publish/unpublish events.
- Announcements/updates appear on website and email notifications work (preferred channel).
- Highlights and resources upload/display function correctly.
- QR attendance marking works during events and admin stats are generated.
- Feedback collection and admin analytics are functional.